# Seyoung Nam

Falls Church, Virginia • +1-206-209-9080 • seyoung.nam@gmail.com • linkedin.com/in/seyoungnam


Infrastructure Software Engineer with 3+ years of experience designing, building, and scaling large-scale distributed systems for cloud infrastructure and traffic engineering. Proven expertise in architecting and optimizing core components like API Gateways and service meshes on k8s to enhance performance, reliability, and security. Passionate about building robust automation and observability solutions that improve developer velocity and reduce operational overhead in high-traffic environments.

## WORK EXPERIENCE

**CISCO(Splunk)** | 7900 Tysons One Place, 10th Fl, Suite #1000, McLean, VA 08/15/2022 – Present

Software Engineer, Traffic Engineering

API Gateway & Ingress Control

- Architected and developed a Kubernetes controller to fully automate ingress routing configurations, reducing new service onboarding time from days to minutes and accelerating feature delivery for product teams
- Engineered and integrated a distributed rate-limiting system with an independent quota service, optimizing resource allocation and reducing monthly AWS operational costs by $18,000
- Improved system performance by implementing zone-aware load balancing for ingress components, significantly decreasing pod-to-pod network latency for high-volume traffic flows
- Increased system reliability and fault tolerance by designing and deploying a multi-zone ingress architecture, ensuring high availability during single-zone failures

Service Mesh & Security (Istio)

- Strengthened the security posture for services by enforcing mutual TLS, ensuring encrypted communication
- Reduced operational burden by streamlining Istio version upgrades using revision-based, canary-style deployments, eliminating service teams' manual intervention
- Enhanced Istio observability by deploying a monitoring service, providing clients with diagrams and metrics for faster problem detection

Observability & Developer Efficiency

- Designed and built a comprehensive observability stack using Prometheus and Grafana to monitor key ingress metrics (RPS, latency, error rates), providing real-time visibility that expedites detection for production issues
- Authored and maintained a knowledge base of over 50 technical articles on core infrastructure(Istio, API Gateway, DNS, and connectivity), which standardized troubleshooting practices and reduced debugging efforts

## PROJECT

Splunk For Health[github] Splunk, C++, Arduino modules/sensors

- Developed a home automation IoT system using Arduino sensors to collect environmental data
- Built a real-time monitoring dashboard in Splunk to visualize data and trigger automated actions based on predefined thresholds

## TECHNICAL SKILLS

Programming Languages: Go, Python, Java, JavaScript, Bash
Infrastructure & Cloud: Kubernetes, Docker, Terraform, GitLab CI/CD, AWS, Azure, GCP
Networking & Service Mesh: Envoy, Istio, gRPC, REST APIs, powerDNS
Database & Monitoring: Prometheus, Grafana, Splunk, PostgreSQL, Redis

## EDUCATION

University of Washington, Seattle, WA Sep 2020 – Jun 2022
Master of Science – Information Management

Peking University, Beijing, China Sep 2007 – Jun 2013
Bachelor of Science – Business Administration